Agreement Among Explainability Methods: A Cross-Model, Cross-Domain Study

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Feature-attribution methods such as SHAP, LIME, and permutation importance are routinely used to justify machine learning decisions, on the implicit assumption that they largely agree on which features matter. We test that assumption directly: across four tabular datasets covering three domains (two educational, one financial, one agricultural), three model families (random forest, gradient boosting, logistic regression), and ten cross-validation folds, we compute 360 feature-importance rankings and compare them pairwise using Kendall's τ along three axes: between folds of the same method, between methods applied to the same model, and between models explained by the same method. Agreement between folds of the same method is the highest of the three (mean τ between 0.54 and 0.89, about 0.75 or higher for three of the four datasets), agreement between different explanation methods applied to the same model is the lowest (mean τ between 0.28 and 0.47), and model-to-model agreement for a fixed method falls in between (0.50 to 0.55); the same order holds when one-hot dummies are grouped back into their original variables. Agreement drops further still (e.g., $\tau = 0.055$ on German Credit) when restricted to the top-10 features actually reported to stakeholders. A two-way ANOVA finds the dataset effect, the comparison-axis effect, and their interaction significant, also when correlated pairwise comparisons are collapsed to cluster means ($p < 10^{-4}$ for all three). Which explanation method is chosen matters more than which model is explained, and a single attribution method, reported without checking it against another, can give a different picture of which features a model relies on.

Povzetek: Metode za razlago napovedi modelov strojnega učenja, kot so SHAP, LIME in permutacijska pomembnost, se pogosto uporabljajo ob predpostavki, da se njihove ocene pomembnosti atributov medsebojno ujemajo. V tej empirični študiji na štirih tabelaričnih naborih podatkov, treh družinah modelov in desetih rezinah prečnega preverjanja neposredno merimo skladnost med metodami s pomočjo Kendallovega koeficienta τ. Rezultati kažejo, da je stabilnost posamezne metode visoka, medtem ko je ujemanje med različnimi metodami na istem modelu bistveno nižje, kar poudarja potrebo po hkratni uporabi več metod razložljivosti v praksi.

1 Introduction

Attribution-based explainability methods are increasingly used as a substitute for genuine model transparency: rather than build inherently interpretable models, practitioners train a black box and then ask a post-hoc method (most commonly SHAP, LIME, or permutation importance) to say

which features drove a prediction. The implicit premise behind this practice is that these methods largely converge on the same answer, so that the choice of which one to run is a matter of convenience rather than substance. That premise is rarely tested; most applied papers report the output of a single method without checking whether a different, equally legitimate method would have told a different story.

We test the premise directly, and deliberately across dissimilar domains, so that any inconsistency we find cannot be dismissed as an artifact of one dataset's idiosyncrasies. We ask three questions. First, how stable is a single explanation method across cross-validation folds of the same model (folds share most of their training data with each other under k-fold CV, so this stability is an upper bound on what independent samples would give, not a fully independent replication): how much of the ranking is signal versus resampling noise? Second, holding the model fixed, how much do SHAP, LIME, and permutation importance agree with each other? Third, holding the explanation method fixed, how much does the ranking change when the underlying model changes? If explanation methods were interchangeable, the second quantity should be close to the first. We show that it is not.

The contributions of this work are: (1) a controlled, quantitative measurement of explanation-method disagreement across four datasets from unrelated domains, three model families, and ten cross-validation folds, decomposed into fold-stability, method-disagreement, and model-disagreement axes; (2) evidence that the ordering of the three axes (folds above models above methods) holds in all four domains, while a factorial ANOVA shows that its magnitude varies by dataset (significant dataset and interaction effects); (3) a practical demonstration, via top-10 feature-ranking comparisons, that this statistical disagreement translates into visibly different feature lists depending on which attribution method produced them.

2 Related Work

SHAP [1] and LIME [2] are the two dominant post-hoc attribution methods in applied machine learning, each grounded in a different theoretical framework (Shapley-value game-theoretic allocation for SHAP, local linear surrogate models for LIME) that gives no a priori guarantee the two will agree in practice. Permutation importance, the third method we use, was formalized as model reliance by Fisher, Rudin and Dominici [3], who additionally showed that importance rankings can vary substantially across different well-performing models fit to the same data (their "Rashomon set" framing), a result that anticipates, from a different angle, our between-model comparison axis. The theoretical plurality of these frameworks is not accidental: Lipton [4] argues that "interpretability" has no consensus definition, and distinguishes transparency (at the level of the whole model, its components, and the training algorithm: simulatability, decomposability, and algorithmic transparency) from post-hoc explanation as competing notions; from this vantage point, SHAP, LIME, and permutation importance each measure a genuinely different thing, and disagreement between them is not a bug to be corrected but a structural fact about the methods.

Most directly relevant, Krishna et al. [5] name and formalize what they call the disagreement problem in explainable machine learning. They interview 25 practising data scientists to define disagreement, propose six metrics for it (feature, rank, sign, and signed-rank agreement over the top-k features, rank correlation, and pairwise rank agreement), and measure it empirically: on two tabular datasets (COMPAS and German Credit), a text and an image dataset, with six explanation methods (LIME, KernelSHAP, and four gradient-based methods) and four tabular models (logistic regression, a neural network, random forest, and gradient-boosted trees), they find that methods often disagree; a user study with another 25 data scientists shows that conflicts are resolved through ad hoc heuristics. Their comparison is between local explanations of individual predictions. Ours compares global rankings and differs in what it adds: permutation importance as a third, global method; the decomposition of agreement into fold stability, method disagreement, and model disagreement under cross-validation; four tabular domains; a repeated-seed control; and a factorial ANOVA. Because their metrics are computed per prediction and ours on

global rankings, our τ values for German Credit are not directly comparable with theirs, although both studies find substantial disagreement on that dataset. An earlier and complementary line of work by Alvarez-Melis and Jaakkola [6] showed that LIME and SHAP are individually brittle to small perturbations of the input, and proposed local sensitivity metrics to quantify this; our repeated-seed control (Results) addresses a different, weaker form of instability, namely sampling noise within a fixed input, and shows it does not account for the between-method disagreement we report. The disagreement we measure, then, is not reducible to per-method noise: it reflects the fact that each method is recovering a structurally different concept of "importance." Doshi-Velez and Kim [7] call for a rigorous science of interpretability, with explicit evaluation of explanations rather than qualitative assessment alone; applied to attribution rankings, this implies that a ranking presented without a concordance check is not validated in the same sense that a predictive model cross-validated against held-out data is validated. Our τ-based concordance measurement is one concrete operationalisation of that principle.

Empirical applications of explanation methods in the literature, meanwhile, typically report a single method's output without checking it against an alternative. In the recent literature of this journal, post-hoc attribution and explainable representations are actively applied across critical domains: Li [8] combines XGBoost and SHAP on 916,000 records of teaching-behaviour data to identify the factors that drive a teaching decision-support model; Yao and Zhu [9] use Random Forest and XGBoost with SHAP to identify cost drivers (such as smoking status, body-mass index, and age) in public-hospital budget prediction; and Vlahek [10] develops an iterative algorithm for explainable feature learning that balances predictive accuracy and interpretability. The two SHAP-based applications report the output of a single attribution method without checking whether another method (such as LIME or permutation importance) would rank the same features first. The same practice appears beyond tabular data: a recent explainable ECG arrhythmia classifier in this journal relies on saliency maps as its interpretability method [11]. Our study directly complements this literature by providing a systematic, quantitative assessment of inter-method concordance across these exact model families and domain structures. A systematic reference for the methods themselves, covering SHAP, LIME, and permutation importance, is provided by Molnar [12]; the aggregation we use to turn LIME's local explanations into a global ranking is described in the Methodology.

The reliability concerns motivating this study are not hypothetical. Slack et al. [13] demonstrate that an adversary can construct a classifier whose biased predictions are invisible to LIME and SHAP by exploiting the perturbation-based local sampling they rely on; if methods can be systematically misled in adversarial conditions, ordinary practitioners operating in good faith face the subtler but structurally analogous risk that two methods evaluated on the same instance will report different rankings for non-adversarial reasons. Kindermans et al. [14] show that widely used saliency methods can change their attributions under input transformations that do not change the model (e.g., a constant shift of the input), so an explanation can be unreliable even when the model is not; our Kendall's τ measurements across 360 rankings quantify a related, cross-method form of unreliability on tabular data. Nauta et al. [15] review the evaluation practices of more than 300 papers that introduce XAI methods, organised by twelve conceptual properties of explanation quality (the Co-12 properties, such as correctness and compactness), and find that one in three of those papers evaluates explanations only with anecdotal evidence; our τ-based comparison is a quantitative evaluation of the kind they call for. Visani et al. [16] quantify this directly by introducing statistical stability indices for LIME, showing that running LIME repeatedly under the same conditions can yield different explanations; we evaluate this in the Results section with a repeated-seed check across five sampling seeds, where same-method agreement is high (mean $\tau = 0.97$ for LIME and 0.91 for permutation importance; lowest for OULAD permutation importance, 0.66) compared with the between-method agreement. Rudin [17] argues, from a theoretical standpoint, that no post-hoc approximation of a black-box model can be fully faithful, and therefore any explanation is, by construction, a partially incorrect

summary of the model; our empirical observation that LIME, SHAP, and permutation importance disagree is consistent with that argument. Speith [18] reviews recent approaches to building taxonomies of XAI methods and argues that no single taxonomy is sufficient to chart the field; under the common distinctions, SHAP, LIME, and permutation importance are all post-hoc and model-agnostic, with SHAP and LIME producing local explanations and permutation importance a global one. This paper compares their outputs empirically on the same models.

3 Methodology

3.1 Datasets

We use four public tabular datasets (characteristics in Table 1), chosen for domain diversity so that a finding replicated across all four cannot be attributed to a property of a single domain: the Open University Learning Analytics Dataset (OULAD, education, classification of the final result into four classes: pass, fail, withdrawn, and distinction), the UCI "Predict Students' Dropout and Academic Success" dataset (education, a different institutional context, three classes: dropout, enrolled, and graduate), the Statlog German Credit dataset (finance, binary), and the Rice (Cammeo and Osmancik) dataset (agriculture, binary classification by grain morphology). All four datasets retain their original class structure: we deliberately did not collapse the multi-class targets to binary, so that the multiclass attribution-aggregation procedure described below is itself exercised and reported, rather than defined away.

3.2 Models and attribution methods

For each dataset we train three models, random forest [19], gradient boosting (XGBoost) [20], and logistic regression, under 10-fold stratified cross-validation, using identical preprocessing (median/most-frequent imputation, standard scaling for numeric features, one-hot encoding for categorical features) across all three so that differences in attribution cannot be attributed to preprocessing differences. The random forest has 100 trees with a maximum depth of 12 (uncapped trees made SHAP computation intractable on OULAD without a gain in accuracy); XGBoost has 100 trees of maximum depth 6 with its default learning rate; logistic regression uses the default L2 penalty with up to 1,000 iterations; all models use the same random seed. For each (dataset, model, fold), we compute three feature-attribution rankings: SHAP (TreeExplainer for tree models, LinearExplainer for logistic regression), LIME (100 random test instances explained locally, with the absolute weights of each feature summed over instances to build one global ranking per fold), and permutation importance [3] (five permutations per feature on a random subsample of at most 2,000 test instances, the same cap applied to SHAP). For multiclass datasets, SHAP and permutation attributions are aggregated across classes by averaging the absolute per-class attribution before ranking; for LIME, each instance's explanation is taken with respect to its own predicted class before aggregation. This yields 3 models $\times$ 3 methods $\times$ 10 folds $\times$ 4 datasets $=$ 360 rankings.

3.3 Predictive performance

Because disagreement between explanation methods is only interesting relative to how much the explained model itself has actually learned, we report each model's held-out predictive performance under the same 10-fold cross-validation used for the attribution comparisons (Table 2). Performance is above both chance ($1/n_{\text{classes}}$) and the majority-class baseline on all four datasets, but by very different margins: models on Rice are strong (accuracy > 0.91, majority baseline 0.57) and Dropout is clearly informative (0.76–0.78, majority 0.50), whereas German Credit (0.75–0.77) is only 5 to 7 points above its majority baseline of 0.70, and OULAD (0.44–0.45 on four classes) is only 6 to 7 points above its majority baseline of 0.38 (chance $= 0.25$) despite being the largest dataset by sample size. Model choice barely affects performance within a dataset (accuracy spans 1.0–1.4 percentage points across the three models on every dataset), meaning the between-model τ axis in Table 3 compares explanation rankings across models of essentially equivalent predictive quality, not a strong model against a weak one. OULAD's comparatively weak absolute performance is worth keeping in mind when interpreting that dataset's attribu-

Table 1: Characteristics of the four datasets. Predictors are given as original variables and, in parentheses, as features after one-hot encoding of the categorical ones, which is the count seen by each model.

Dataset	Domain	Task	n	Predictors	Class balance
OULAD	Education	4-class	32,593	10 (49)	Pass 37.9% / Withdrawn 31.2% / Fail 21.6% / Distinction 9.3%
Dropout	Education	3-class	4,424	36	Graduate 49.9% / Dropout 32.1% / Enrolled 17.9%
German Credit	Finance	Binary	1,000	20 (61)	Good risk 70.0% / Bad risk 30.0%
Rice	Agriculture	Binary	3,810	7	Osmancik 57.2% / Cammeo 42.8%

tion rankings: a model that predicts only modestly better than chance has less genuine signal for any explanation method to recover, so some of OULAD's between-method disagreement could reflect the model's own limited signal rather than a property of the explanation methods themselves; the same caution applies, to a lesser degree, to German Credit, the dataset with the lowest top-10 agreement.

3.4 Agreement metric

We measure pairwise ranking agreement with Kendall's τ and organize comparisons along three axes: *between folds* (same model, same method, different fold; a measure of the method's own stability), *between methods* (same model, same fold, different method; the central quantity of interest), and *between models* (same method, same fold, different model; whether an explanation method's conclusions generalize across model families). For each axis we report the mean and median τ with a 2,000-resample bootstrap 95% confidence interval, and we fit a two-way ANOVA of τ on dataset and comparison axis (with interaction) to test whether the pattern is consistent across datasets. The 2,340 pairwise τ values feeding that ANOVA are not fully independent: the 405 between-fold pairs per dataset, for example, are built from only 10 underlying fold rankings per (model, method), so as a robustness check we repeat the ANOVA on cluster means. A cluster is one (model, method) combination on the between-folds axis (9 per dataset), one (model, fold) combination on the between-methods axis (30), and one (method, fold) combination on the between-models axis (30), which collapses 2,340 values to $4 \times 69 = 276$. This reduces pseudoreplication but does not remove it, since clusters on different axes still share the same underlying rankings; the axis, dataset, and interaction effects remain significant under this specification (Results).

4 Results

The pattern in Table 3 and Figures 1, 2, and 3 is consistent across all four datasets: stability across folds of the *same* method is high (mean τ from 0.54 to 0.89), agreement *between* methods applied to the same model is roughly half that (mean τ from 0.28 to 0.47), and agreement between models for a fixed method sits in between (0.50 to 0.55). A two-way ANOVA of τ on dataset and comparison axis (Table 4) finds both main effects significant (dataset: $F = 222.6$, $p < 10^{-126}$, partial $\eta^2 = 0.223$; axis: $F = 527.1$, $p < 10^{-188}$, partial $\eta^2 = 0.312$) with a significant but comparatively small interaction ($F = 18.9$, $p < 10^{-20}$, partial $\eta^2 = 0.047$), meaning the magnitude of the axis differences depends on the dataset while their order is the same in all four (Table 3), and the comparison axis (fold/method/model) accounts for a somewhat larger share of variance in τ than the dataset does. Because the raw τ values are clustered (many pairs share the same underlying fold rankings), we repeat this test on the cluster-mean specification described in Methodology; both main effects and the interaction remain significant (dataset: $F = 22.5$, $p < 10^{-12}$; axis: $F = 122.4$, $p < 10^{-37}$; interaction: $F = 5.0$, $p < 10^{-4}$), so the pattern is not only an artifact of treating correlated pairwise comparisons as independent observations. The between-method agreement for the Rice dataset has the widest confidence interval ($[0.40, 0.53]$, Table 3), and some of its individual between-method τ values are nega-

Table 2: Held-out predictive performance (mean ± SD across 10 folds), same cross-validation split used throughout.

Dataset	Model	Accuracy	F1 (macro)	Log-loss
OULAD	Random forest	0.446 ± 0.006	0.286 ± 0.006	1.212 ± 0.005
	Gradient boosting	0.440 ± 0.006	0.320 ± 0.005	1.223 ± 0.009
	Logistic regression	0.450 ± 0.010	0.307 ± 0.009	1.205 ± 0.006
Dropout	Random forest	0.775 ± 0.016	0.692 ± 0.023	0.572 ± 0.022
	Gradient boosting	0.777 ± 0.017	0.710 ± 0.021	0.616 ± 0.058
	Logistic regression	0.763 ± 0.013	0.673 ± 0.022	0.577 ± 0.024
German Credit	Random forest	0.758 ± 0.033	0.656 ± 0.049	0.492 ± 0.028
	Gradient boosting	0.765 ± 0.024	0.698 ± 0.041	0.620 ± 0.078
	Logistic regression	0.751 ± 0.029	0.680 ± 0.040	0.502 ± 0.043
Rice	Random forest	0.927 ± 0.013	0.925 ± 0.013	0.246 ± 0.058
	Gradient boosting	0.916 ± 0.010	0.914 ± 0.010	0.251 ± 0.035
	Logistic regression	0.929 ± 0.012	0.928 ± 0.012	0.183 ± 0.017

Table 3: Mean Kendall's τ by comparison axis and dataset (bootstrap 95% CI in brackets).

Dataset	Between folds	Between methods	Between models
OULAD	0.75 [0.73, 0.76]	0.46 [0.44, 0.48]	0.55 [0.52, 0.57]
Dropout	0.76 [0.75, 0.78]	0.41 [0.38, 0.45]	0.55 [0.53, 0.57]
German Credit	0.54 [0.51, 0.57]	0.28 [0.26, 0.31]	0.50 [0.47, 0.53]
Rice	0.89 [0.88, 0.91]	0.47 [0.40, 0.53]	0.55 [0.50, 0.60]

tive (Figure 2); with only 7 features, τ is volatile and a few swapped positions can push it below zero. Rice's morphological features are also strongly correlated with each other (area, convex area, and perimeter all measure grain size), and SHAP and permutation importance can distribute credit among correlated features differently, a known source of disagreement that we did not control for. By method pair, SHAP and permutation importance agree most with each other on three datasets (mean between-method τ 0.84 on Rice, 0.61 on Dropout, 0.55 on OULAD), and LIME accounts for most of the disagreement there; on German Credit the closest pair is SHAP and LIME (0.43).

The number of features per dataset ranges from 7 (Rice) to 61 (German Credit, post-encoding), and τ on the full ranked list could misrepresent agreement on the features that matter, since most ranked positions in the higher-dimensional datasets are low-importance features whose relative order is close to noise. To check this, we recomputed between-method τ restricted to the union of each pair's top-10 features (the portion of the ranking a stakeholder reads), keeping each feature's position in the full ranking, so a feature in one method's top 10 enters the comparison with its full-list rank under the other method. Restricting to the top 10 *reduces* agreement for the three higher-dimensional datasets (dropout: 0.414 full-list vs. 0.244 top-10; German Credit: 0.284 vs. 0.055; OULAD: 0.461 vs. 0.189), while Rice, whose full feature list is already close to 10, is essentially unchanged (0.467 vs. 0.467). The full-list τ is therefore raised by agreement in the tail of the ranking: on the features most likely to be reported to a stakeholder, agreement among SHAP, LIME, and permutation importance is lower than the values in Table 3, for three of the four datasets.

Because LIME and permutation importance are both estimated from a random subsample of test instances and not computed exactly, part of the between-method disagreement we report could in principle be estimator noise rather than genuine disagreement between the underlying methods. We tested this directly with a repeated-seed control: holding the trained model, dataset,

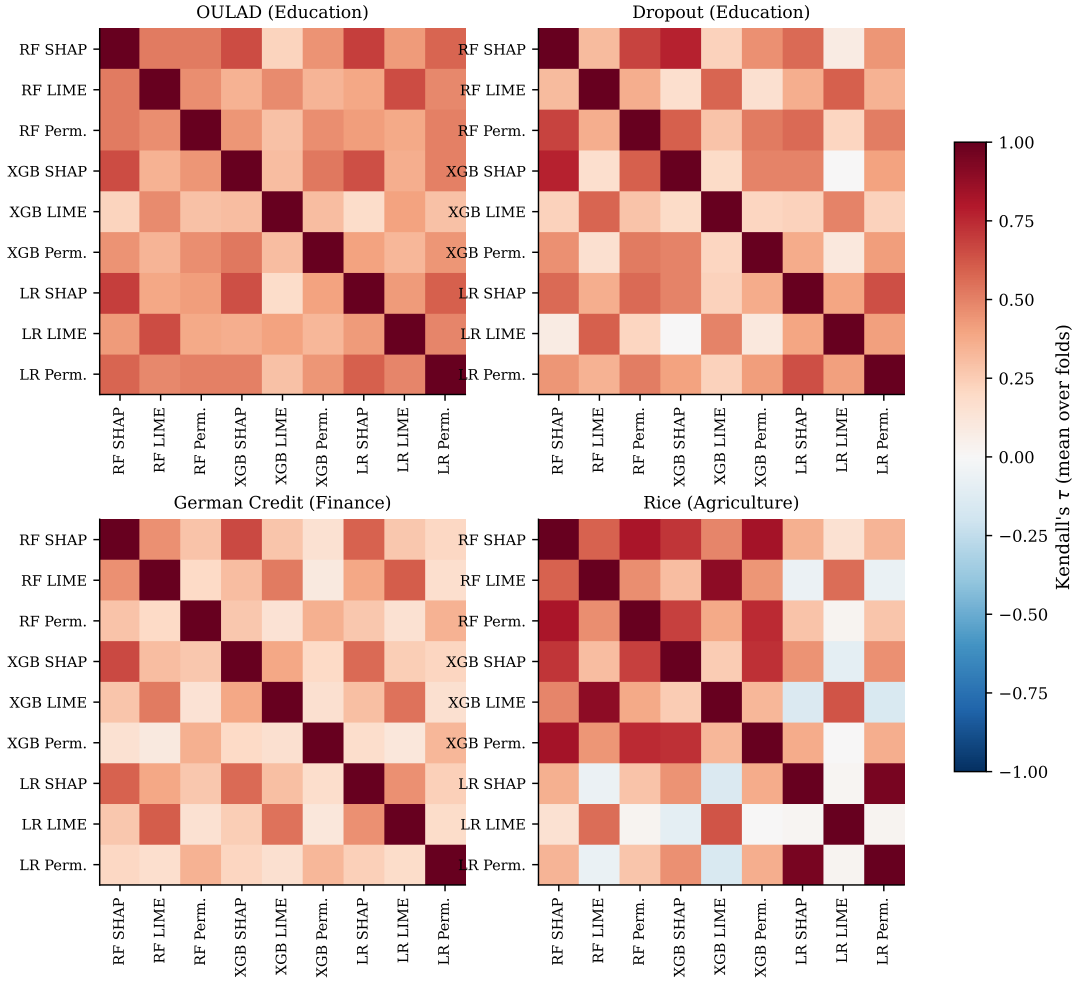


Figure 1: Pairwise Kendall's τ concordance (mean over folds) among the nine (model, method) combinations, by dataset; RF = random forest, XGB = gradient boosting, LR = logistic regression, Perm. = permutation importance.

and fold fixed, we recomputed each method's ranking five times with different sampling seeds and measured τ between repetitions of the *same* method. If between-method τ were mostly estimator noise, this same-method τ should be comparably low. (This control covers one fold per dataset and model.) Instead it is high: mean $\tau = 0.970$ for LIME and 0.915 for permutation importance across the four datasets (only OULAD's permutation-importance repetitions show any real spread, mean $\tau = 0.660$; all other dataset-method combinations reach at least 0.94). Estimator noise is therefore a minor contributor to the reported between-method disagreement (mean $\tau \in [0.28, 0.47]$): re-running the same method with a different seed reproduces essentially the same ranking, while switching methods does not.

With one-hot encoding, each category of a categorical variable is ranked as a separate feature (for example `chk_status_A14`), and permuting a single dummy creates rows that cannot occur, which can inflate disagreement in the two datasets with many dummies. We therefore summed each method's attributions over the dummies of each original variable and recomputed the three axes (for permutation importance this sums the per-dummy importances, an approximation to permuting the whole variable). Agreement rises on both datasets: between methods from 0.28 to 0.40 on German Credit and from 0.46 to 0.65 on OULAD, between folds to 0.64 and 0.91, and between models to 0.59 and 0.78. Part of the disagreement on these two datasets is therefore due

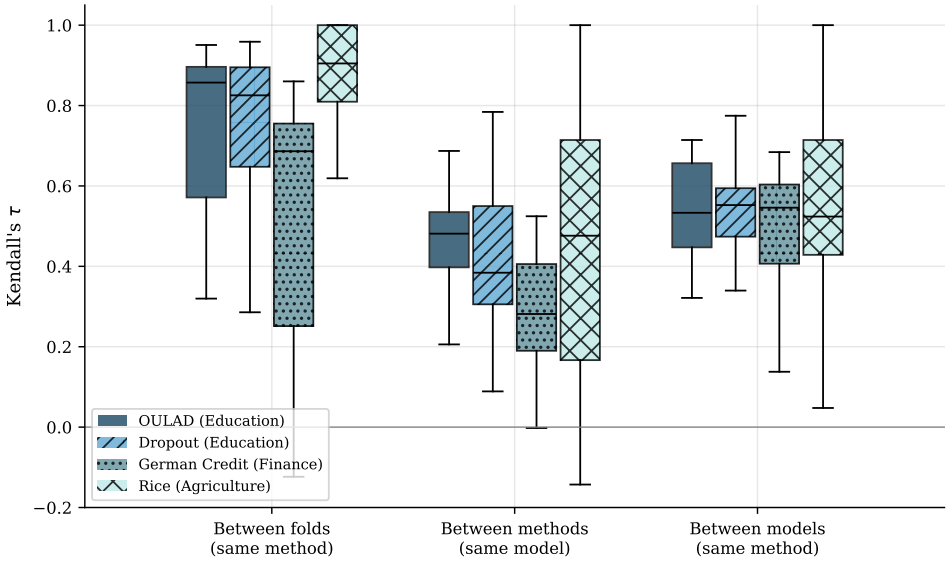


Figure 2: Kendall's τ by comparison axis and dataset.

to ranking dummies separately, but the order of the three axes is unchanged, and between-method agreement remains the lowest.

In practical terms, if two researchers explain the same trained random forest, one with SHAP and one with LIME, their rankings of which features matter will typically show 64%–74% of feature pairs in the same relative order (Kendall's $\tau \approx 0.28$–0.47, corresponding to $(\tau+1)/2$ concordant pairs), while the same researcher re-running the same method on a different resampling fold of the same data will get a ranking that agrees with their first run substantially more (77%–95% concordant pairs, $\tau \approx 0.54$–0.89, and about 0.75 or higher for three of the four datasets). Choice of model family matters less than choice of explanation method: switching models while keeping the explanation method fixed gives $\tau \approx 0.50$–0.55, higher than the $\tau \approx 0.28$–0.47 obtained by switching explanation methods while keeping the model fixed, and both fall below the within-method stability across folds.

Figure 4 makes the disagreement concrete rather than only statistical: for the random forest model, several features that rank in SHAP's top 10 do not appear in LIME's top 10 at all (no connecting line), and the ordering among the features the two methods do share still differs. This is the practical face of the $\tau \approx 0.28$–0.47 figure above: a feature list handed to a stakeholder would visibly change depending on which method produced it.

5 Discussion

The consistency of this pattern across four unrelated domains (two educational datasets with different institutional contexts, a financial credit dataset, and an agricultural morphology dataset) argues against explaining it away as a quirk of one problem. A conclusion of the form "feature X is the most important driver of this model's predictions," when it rests on a single attribution method, is method-contingent: a different, equally standard method applied to the identical trained model shows top-10-restricted concordance between $\tau = 0.055$ (German Credit) and $\tau = 0.244$ (Dropout) for three of the four datasets (Results), meaning the specific features flagged as most important frequently differ between methods.

This matters most in exactly the settings where attribution methods are invoked as justification (credit decisions, admissions or dropout-risk flagging, medical triage) because the audience for such an explanation (a regulator, an affected individual, an auditor) has no way to know that the reported ranking is method-contingent unless the analyst says so. Our recommendation, consistent with what the agreement numbers support, is to report at least two attribution methods and their concordance (a single τ value is cheap to compute) rather than a single ranking presented as though it were the model's ground truth.

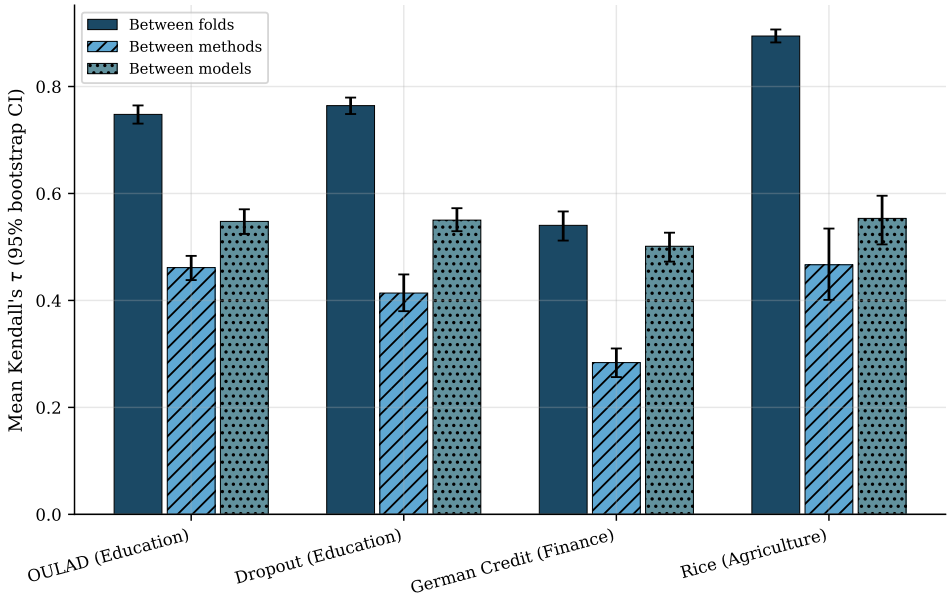


Figure 3: Mean Kendall's τ by dataset and comparison axis (95% bootstrap CI): the order between folds > between models > between methods holds in all four domains, with dataset-dependent magnitudes.

Table 4: Two-way ANOVA of Kendall's τ on dataset and comparison axis, raw pairwise specification ($n = 2{,}340$). Partial $\eta^2 = SS_{\text{effect}}/(SS_{\text{effect}} + SS_{\text{residual}})$.

Source	Sum of squares	df	F	p	Partial η^2
Dataset	24.06	3	222.63	5.7×10^{-127}	0.223
Axis	37.98	2	527.12	1.5×10^{-189}	0.312
Dataset × Axis	4.10	6	18.95	1.2×10^{-21}	0.047
Residual	83.88	2,328			

The comparatively higher between-model agreement points to a specific mechanism: it suggests that a chosen explanation method captures something at least partially about the *data*, namely which features are genuinely informative for the task, that persists somewhat across model families, whereas the low between-method agreement suggests each method also imposes its own distinct lens (SHAP's game-theoretic allocation, LIME's local linear surrogate, permutation's direct ablation) that shapes the ranking as much as the underlying data does.

Our results agree with Krishna et al. [5], who find frequent disagreement among local explanations and report that practitioners resolve it with ad hoc heuristics; we add that the disagreement persists in global rankings, in four tabular domains, and beyond what fold resampling and estimator noise produce. We do not have a ground truth for which features the models truly rely on, so we cannot say which method is closer to it; a synthetic dataset with known relevant features would be needed for that.

6 Limitations

Two of the four datasets are only modestly predictable: OULAD and German Credit models exceed their majority-class baselines by just 5 to 7 accuracy points (Section 3.3), so their attribution rankings explain models with limited signal, and their low agreement may partly reflect that. This study is limited to tabular data; image and text models, where SHAP and LIME operate over different units (pixels/superpixels, tokens) rather than a shared feature space, were out of scope. We compare three widely used methods out of a

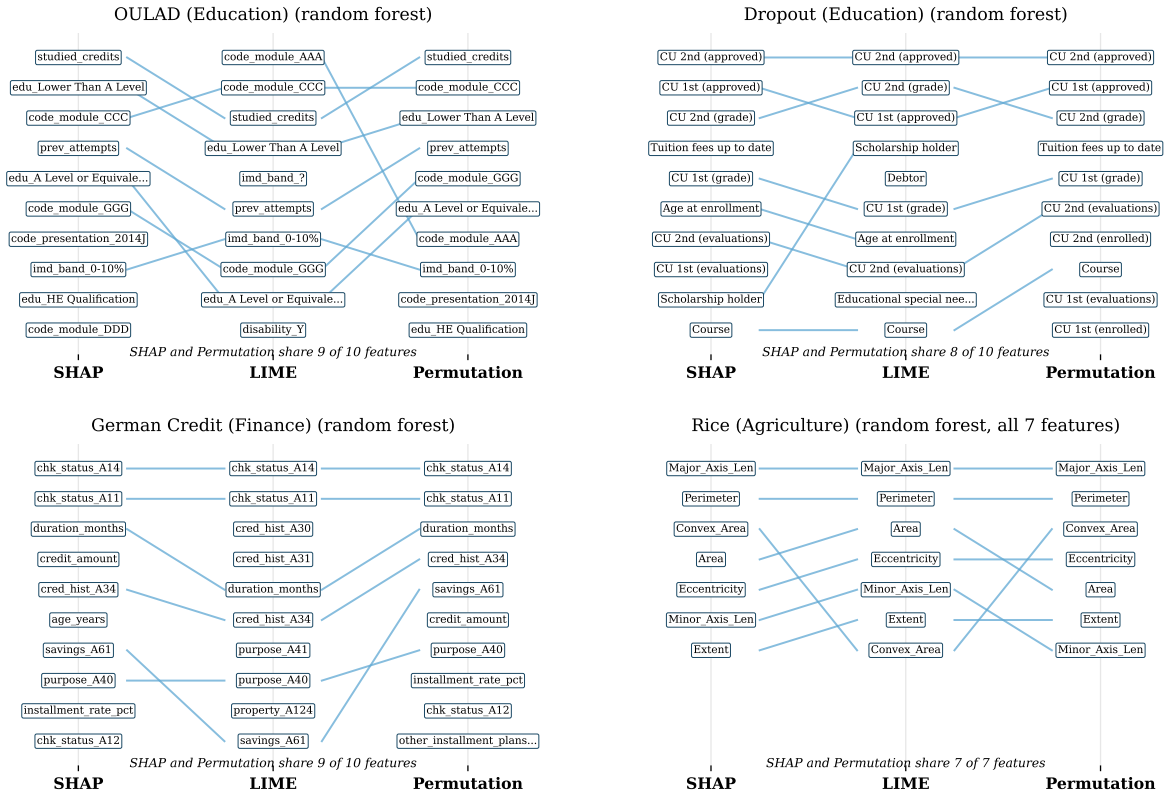


Figure 4: Top-10 features by SHAP, LIME, and permutation importance (random forest), all four datasets (Rice has only 7 features); lines connect a feature that appears in the top 10 of both adjacent methods, and the note under each panel gives the overlap between SHAP and permutation importance, which are not adjacent. German Credit categories keep their UCI codes (e.g., `chk_status_A14`: no checking account).

substantially larger family (e.g., Integrated Gradients, counterfactual explanations, anchors were not included). LIME's aggregation from local, per-instance explanations to a single global ranking is itself a design choice with more than one reasonable implementation; we used one documented, defensible approach, but did not exhaustively test alternative aggregation schemes. Furthermore, LIME discretizes continuous features into quantiles by default to stabilize linear surrogate fitting, whereas SHAP and permutation importance evaluate features in continuous space; while this follows standard applied practice, part of the between-method divergence reflects this structural difference in perturbation geometry.

In multiclass settings (OULAD and Dropout), SHAP aggregates attribution across all candidate classes, whereas LIME conditions on the predicted class; however, on strictly binary tasks (German Credit and Rice), where this outcome asymmetry does not exist, inter-method agreement remains equally suppressed ($\tau \in [0.28, 0.47]$), suggesting that the disagreement is not an artifact of class-aggregation asymmetry. Preprocessing (standardization and imputation) was applied identically to all models and folds, and tree-based ensembles are invariant to monotonic scaling, so preprocessing differences cannot account for the inter-method disagreement. Two choices do affect it: ranking one-hot dummies separately, which lowers agreement on German Credit and OULAD (Results), and correlated features, among which methods may distribute credit differently.

The between-model axis additionally confounds model family with explainer algorithm, since SHAP uses `TreeExplainer` for the tree-based models and `LinearExplainer` for logistic regression; disagreement attributed to "model" on that axis may partly reflect this estimator change. We checked this directly: if the explainer swap were inflating SHAP's between-model disagreement, SHAP should show the *lowest* between-model τ of the three methods, since it is the only one whose estimator changes across models. It shows the highest instead (mean between-model τ: SHAP = 0.595, LIME = 0.580, permutation importance = 0.439, pooled across datasets and folds), which runs opposite to what the confound would predict and is more consistent with permutation importance, not SHAP, being the outlier on this axis for reasons unrelated to the explainer-swap concern. This does not rule out every possible effect of the estimator change, but it argues against it being a major driver of the reported between-model figures.

7 Conclusion

Across four datasets from unrelated domains, three model families, and ten cross-validation folds, feature-attribution methods that are routinely treated as interchangeable diagnostic tools produce rankings with mean $\tau \in [0.28, 0.47]$ against each other (64%–74% concordant feature pairs), consistently lower than their agreement with themselves across resampling folds ($\tau \in [0.54, 0.89]$, 77%–95% concordant pairs), and lower still when the comparison is restricted to the top-10 features most likely to be reported to a stakeholder. Choice of explanation method has a measurable effect on the resulting ranking, of a magnitude that a repeated-seed control shows is not attributable to estimator noise, and that remains when one-hot dummies are grouped into their original variables. We recommend reporting concordance between at least two attribution methods, rather than the output of a single one, for explainability claims used to justify decisions.

Data Availability

The preprocessed datasets, trained models, attribution rankings, and analysis code are permanently archived and openly accessible in Zenodo at https://doi.org/10.5281/zenodo.22907629 and GitHub at https://github.com/JuanAvilesP3/informatica-concordancia-xai.

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Conflict of Interest

The authors declare that they have no conflict of interest.

Author Contributions

Italo Javier Tenempaguay-Granizo: Conceptualization, Methodology, Formal Analysis, Writing - Original Draft. **Juan Pablo Aviles-Esparza:** Software, Validation, Investigation, Visualization, Writing - Review & Editing. **Isaac David Torres-Paredes:** Supervision, Project Administration, Resources, Writing - Review & Editing.

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